

Dementias - Objectives

- Recognize clinical signs of cognitive impairment by understanding realms of cognitive function including attention and consciousness, language, memory, visuospatial and executive functions.
- Identify the difference between delirium and dementia, and enumerate some potentially reversible causes.
- For the major neurocognitive disorders (NCDs), including Alzheimer's Disease, Vascular Dementia, Frontotemporal Dementia, Dementia with Lewy Bodies, Parkinson's disease dementia, and others discussed in class:
 - Know the defining clinical features including epidemiologic factors, presenting symptoms, natural history
 - Know the classic findings on history, physical exam, cognitive testing, lab features and imaging that help us diagnose and distinguish among the NCDs.
 - Recognize the psychosocial impacts of cognitive impairments on patients and their families and understand the role of the clinician in their care.
- For the major dementia syndromes, know the underlying pathophysiologic processes including
 - The roles of APP, alpha-, beta-, and gamma secretases, apolipoproteins, tau, cholesterol, the glymphatic system, and lipid rafts in the pathophysiology of Alzheimers disease
 - Recognize and describe senile (amyloid) plaques and neurofibrillary tangles (hyperphosphorylated tau), by appearance on histological slides and by molecular composition.
 - Recognize and describe Lewy bodies, by appearance on histological slides and by molecular composition (alpha-synuclein)
 - Describe the molecular composition of deposits often found in Frontotemporal dementia (tau and ubiquitin) and the role of changes in serotonin neurotransmission in this type of dementia
 - Describe multi-infarct dementia and how "silent stroke" contributes to it.
- For the treatments of dementia,
 - List and describe the non-pharmacological treatment options recommended for caregivers
 - Know the different classes of drugs used to treat dementia, the neurotransmitter systems they affect, and their mechanism of action.
 - Know what treatments are used for different aspects of dementia, including behavioral changes and sleep disturbances.

- Know which drugs belong in each class.
- Know the indications, contraindications, and common or serious adverse reactions particular to each drug or class of drugs.
- Understand how the properties of absorption, distribution, metabolism, excretion and tolerance contribute to the clinical effects of these drugs.